Exam 3 (solved)

Feb 9 2026

Question 1

(a) If a square matrix A has all n of its singular values equal to 1 in the SVD, what basic classes of matrices does A belong to? (Singular, symmetric, orthogonal, positive definite or semidefinite, diagonal)

If all singular values of A are 1, then A is an orthogonal matrix (or unitary, if complex). This is because the SVD of A is $A = U\,\mathrm{diag}(1,\ldots,1)V^T = UV^T$, and U, V are orthogonal matrices, so A is a product of orthogonal matrices, which is itself orthogonal. Orthogonal matrices have all singular values equal to 1, and their inverse is their transpose. A could also be diagonal if $U = V$ and A is diagonal with $\mathrm{diag}(\pm 1,\ldots,\pm 1)$, but the key class is orthogonal.

Step 1: The singular values of A are the entries of $\mathrm{diag}(\sigma_1,\ldots,\sigma_n)$ in the SVD $A = U\,\mathrm{diag}(\sigma_1,\ldots,\sigma_n)V^T$.

Step 2: If all $\sigma_i = 1$, then $A = UV^T$.

Step 3: U and V are orthogonal, so A is orthogonal (since the product of orthogonal matrices is orthogonal).

Step 4: A could also be diagonal if $U = V$ and A is diagonal with all entries ± 1, but the main class is orthogonal.

Conclusion: A is orthogonal (and possibly also diagonal if $U = V$). A is not necessarily symmetric, positive definite, or positive semidefinite, and is not singular.

(b) Suppose the (orthonormal) columns of H are eigenvectors of B:

$$H = \frac{1}{2}[1 \quad 1 \quad -1 \quad -1\, 1 \quad -1 \quad -1 \quad 1\, 1 \quad 1 \quad 1 \quad 1\, 1 \quad -1 \quad 1 \quad -1] \qquad H^{-1} = H^{\mathrm{T}}$$

The eigenvalues of B are $\lambda = 0, 1, 2, 3$. Write B as the product of 3 specific matrices. Write $C = (B+I)^{-1}$ as the product of 3 matrices.

Since H is orthogonal and its columns are eigenvectors of B, B is diagonalizable as $B = H\Lambda H^T$, where $\Lambda = \mathrm{diag}(0,1,2,3)$. This is the spectral decomposition.

Step 1: B is diagonalizable with $B = H\Lambda H^T$, where $\Lambda = \mathrm{diag}(0,1,2,3)$.

Step 2: H is orthogonal, so $H^{-1} = H^T$.

Step 3: B as a product of 3 matrices: $B = H\Lambda H^T$.

Step 4: $C = (B+I)^{-1} = H(\Lambda + I)H^T)^{-1} = H(\Lambda + I)^{-1}H^T$ (since H is orthogonal, $H^T = H^{-1}$ and H commutes with diagonal matrices in this form).

Step 5: $\Lambda + I = \mathrm{diag}(1, 2, 3, 4)$, so $(\Lambda + I)^{-1} = \mathrm{diag}(1, 1/2, 1/3, 1/4)$.

Conclusion:

- $B = H\Lambda H^T$ with $\Lambda = \mathrm{diag}(0, 1, 2, 3)$.
- $C = (B+I)^{-1} = H\mathrm{diag}(1, 1/2, 1/3, 1/4)H^T$.

(c) Using the list in question (a), which basic classes of matrices do B and C belong to? (Separate question for B and C)

B is symmetric (since $B = H\Lambda H^T$ with H orthogonal and Λ diagonal and real), and diagonalizable. B is not orthogonal (its eigenvalues are not all ± 1), not positive definite (it has a zero eigenvalue), and is **singular** (since $\det B = 0$ or equivalently, 0 is an eigenvalue). Furthermore, since all eigenvalues are $0, 1, 2, 3$ (all ≥ 0), B is **positive semidefinite**. B is not diagonal in the standard basis, but is diagonalizable.

C is also symmetric (since $C = HDH^T$ with D diagonal and real), and all its eigenvalues are positive ($1, 1/2, 1/3, 1/4$), so C is positive definite. C is not orthogonal, not diagonal in the standard basis, and not singular.

Step 1: B is symmetric and diagonalizable, but not orthogonal, not positive definite, and not diagonal in the standard basis. B is **singular** and **positive semidefinite**.

Step 2: C is symmetric and positive definite (all eigenvalues > 0), but not orthogonal, not diagonal in the standard basis.

Conclusion:

- B is symmetric, diagonalizable, **singular**, and **positive semidefinite**.
- C is symmetric and positive definite.

Question 2

(a) Find three eigenvalues of A, and an eigenvector matrix S:

Let's find the eigenvalues of

$$A = [-1 \quad 2 \quad 4\,0 \quad 0 \quad 5\,0 \quad 0 \quad 1]$$

Step 1: A is upper triangular, so its eigenvalues are the diagonal entries: $-1, 0, 1$.

Step 2: To find an eigenvector matrix S, we solve $(A-\lambda I)x=0$ for each eigenvalue.

- For $\lambda=-1$: $A+I=[0 \quad 2 \quad 4\,0 \quad 1 \quad 5\,0 \quad 0 \quad 2]$ The first row gives $2x_2+4x_3=0$, the second $x_2+5x_3=0$, the third $2x_3=0$ so $x_3=0$, $x_2=0$, x_1 free. So eigenvector is $[1,0,0]^T$.

- For $\lambda=0$: $A-0I=A$. The third row: $x_3=0$. Second row: $5x_3=0$ so $x_3=0$. First row: $-x_1+2x_2=0$ so $x_1=2x_2$. So eigenvector is $[2,1,0]^T$.

- For $\lambda=1$: $A-I=[-2 \quad 2 \quad 4\,0 \quad -1 \quad 5\,0 \quad 0 \quad 0]$ Third row: $0=0$ (free). Second row: $-y_2+5y_3=0 \implies y_2=5y_3$. First row: $-2y_1+2y_2+4y_3=0$. Substitute $y_2=5y_3$: $-2y_1+10y_3+4y_3=0 \implies -2y_1+14y_3=0 \implies y_1=7y_3$. So eigenvector is $[7,5,1]^T$.

Conclusion: The eigenvalues are $-1,0,1$ and an eigenvector matrix is

$$S=[1 \quad 2 \quad 7\,0 \quad 1 \quad 5\,0 \quad 0 \quad 1]$$

(b) Explain why $A^{1001}=A$. Is $A^{1000}=I$? Find the three diagonal entries of e^{At}.

Because A is upper triangular with eigenvalues $-1,0,1$, and its eigenvector matrix S is invertible, A is diagonalizable: $A=S\Lambda S^{-1}$ with $\Lambda=\operatorname{diag}(-1,0,1)$. Then $A^k=S\Lambda^k S^{-1}$.

Step 1: $\Lambda^k=\operatorname{diag}((-1)^k,0^k,1^k)$. For $k\geq 1$, $0^k=0$.

Step 2: $A^{1001}=S\Lambda^{1001}S^{-1}$. $(-1)^{1001}=-1$, $0^{1001}=0$, $1^{1001}=1$. So $\Lambda^{1001}=\Lambda$.

Step 3: Therefore $A^{1001}=S\Lambda S^{-1}=A$.

Step 4: $A^{1000}=S\Lambda^{1000}S^{-1}$. $(-1)^{1000}=1$, $0^{1000}=0$, $1^{1000}=1$. So $\Lambda^{1000}=\operatorname{diag}(1,0,1)$. This is not the identity matrix, so $A^{1000}\neq I$.

Step 5: The diagonal entries of e^{At} are $e^{-t},1,e^t$, since $e^{At}=Se^{\Lambda t}S^{-1}$ and $e^{\Lambda t}=\operatorname{diag}(e^{-t},1,e^t)$.

Conclusion:

- $A^{1001}=A$.
- $A^{1000}\neq I$.
- The diagonal entries of e^{At} are $e^{-t},1,e^t$.

(c) The matrix $A^{\mathrm{T}}A$ (for the same A) is

$$A^{\mathrm{T}}A=[1 \quad -2 \quad -4-2 \quad 4 \quad 8-4 \quad 8 \quad 42]$$

How many eigenvalues of $A^{\mathrm{T}}A$ are positive? zero? negative? (Don't compute them but explain your answer.) Does $A^{\mathrm{T}}A$ have the same eigenvectors as A?

$A^{\mathrm{T}}A$ is always symmetric and positive semidefinite (see notes on SVD and least squares). Its eigenvalues are the squares of the singular values of A, so they are all ≥ 0.

Step 1: A is 3×3 and has rank 2 (since the second and third rows are linearly independent, but the first row is a combination of the others), so $A^{\mathrm{T}}A$ has rank 2.

Step 2: Therefore, $A^{\mathrm{T}}A$ has two positive eigenvalues and one zero eigenvalue.

Step 3: $A^{\mathrm{T}}A$ cannot have negative eigenvalues because it is positive semidefinite.

Step 4: $A^{\mathrm{T}}A$ does not generally have the same eigenvectors as A (see notes on SVD and diagonalization: the eigenvectors of A and $A^{\mathrm{T}}A$ are generally different unless A is normal or symmetric).

Conclusion:

- $A^{\mathrm{T}}A$ has two positive eigenvalues and one zero eigenvalue; none are negative.
- $A^{\mathrm{T}}A$ does not have the same eigenvectors as A in general.

Question 3

(a) What are the eigenvalues and eigenvectors of A^{-1}? Prove that your answer is correct.

If A has orthonormal eigenvectors $q_1, \ldots, q_n$ and positive eigenvalues $\lambda_1, \ldots, \lambda_n$, then A^{-1} has the same eigenvectors q_j, but its eigenvalues are $1/\lambda_j$. This follows from the definition of eigenvalues and the invertibility of A (since all $\lambda_j > 0$).

1. By definition, $Aq_j = \lambda_j q_j$ for each j.
2. Since A is invertible (all $\lambda_j > 0$), we can apply A^{-1} to both sides: $A^{-1}(Aq_j) = A^{-1}(\lambda_j q_j)$.
3. This gives $q_j = \lambda_j A^{-1} q_j$, so $A^{-1}q_j = (1/\lambda_j)q_j$.
4. Therefore, q_j is an eigenvector of A^{-1} with eigenvalue $1/\lambda_j$.

Therefore the eigenvalues of A^{-1} are $1/\lambda_1, \ldots, 1/\lambda_n$, and the eigenvectors are $q_1, \ldots, q_n$ (the same as for A).

(b) Any vector b is a combination of the eigenvectors: $b = c_1q_1 + \cdots + c_nq_n$. What is a quick formula for c_1 using orthogonality of the q's?

Because the q_j are orthonormal, the coefficient c_1 is simply the dot product $q_1^T b$ (or $q_1 \cdot b$). This is a direct result of the orthonormality property.

1. Write $b = \sum_{j=1}^{n} c_j q_j$.
2. Take the dot product of both sides with q_1: $q_1^T b = \sum_{j=1}^{n} c_j q_1^T q_j$.
3. Since the q_j are orthonormal, $q_1^T q_j = 0$ for $j \neq 1$ and $q_1^T q_1 = 1$.
4. Therefore, $q_1^T b = c_1$.

Therefore $c_1 = q_1^T b$.

(c) The solution to $Ax = b$ is also a combination of the eigenvectors: $A^{-1}b = d_1 q_1 + \cdots + d_n q_n$. What is a quick formula for d_1? You can use the c's even if you didn't answer part (b).

The coefficient d_1 is c_1/λ_1, where c_1 is the coefficient of q_1 in b. This is because A^{-1} acts on each eigenvector by scaling it by $1/\lambda_j$.

1. $b = \sum_{j=1}^{n} c_j q_j$.
2. $A^{-1}b = A^{-1}(\sum_{j=1}^{n} c_j q_j) = \sum_{j=1}^{n} c_j A^{-1} q_j$.
3. From part (a), $A^{-1} q_j = (1/\lambda_j) q_j$.
4. Therefore, $A^{-1}b = \sum_{j=1}^{n} (c_j/\lambda_j) q_j$.
5. So $d_1 = c_1/\lambda_1$.

Therefore $d_1 = c_1/\lambda_1$ (and in general $d_j = c_j/\lambda_j$).